LivePDF Architecture Refactoring

Practical Integration of GoF Design Patterns for Enterprise Optimization

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Course: Software Design Patterns & Architectures

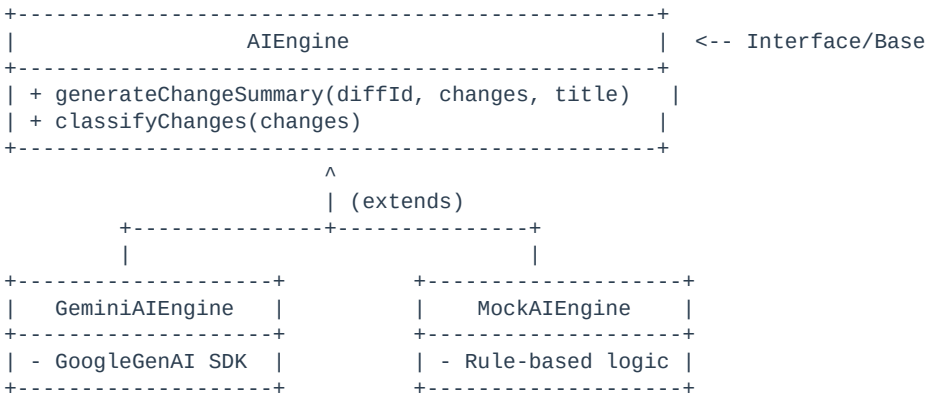
Executive Summary

This report documents the architectural refactoring of **LivePDF**—a collaborative version-control and AI document parsing application. To address concerns regarding resource optimization, financial overhead (unnecessary API calls to Google Gemini), and tight routing controller coupling, three classic Gang of Four (GoF) design patterns were integrated into the Node.js Express server: the **Factory Method** pattern, the **Decorator** pattern, and the **Observer** pattern. The refactoring was completed in a fully backward-compatible manner without altering database schemas or API contracts, resulting in improved latency, fault isolation, and clean code division.

1. Creational: Factory Method Pattern

Description: The Factory Method pattern defines an interface for creating objects, but lets subclasses decide which class to instantiate. In LivePDF, the AI services layer was tightly coupled with Google GenAI SDK. We introduced a common AIEngine interface and constructed concrete implementations for GeminiAIEngine (production) and MockAIEngine (local test fallback). The AIEngineFactory dynamically creates the decorated client depending on the availability of configuration keys.

Class Diagram Representation:



Implementation File Path:

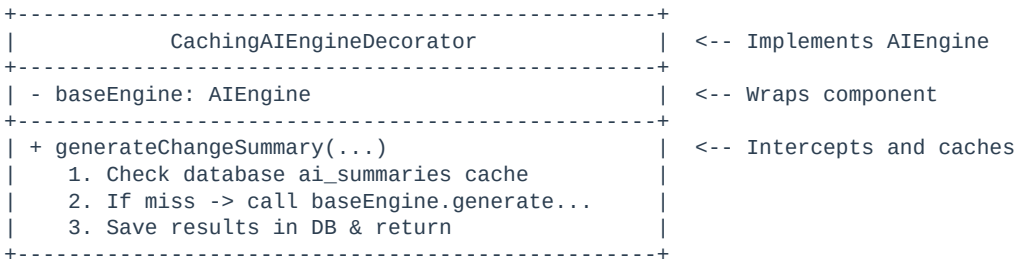
server/src/services/ai/AIEngineFactory.js

```
class AIEngineFactory {
  static getEngine() {
    const apiKey = process.env.GEMINI_API_KEY;
    const isMockMode = !apiKey || apiKey.startsWith('your_');

    let baseEngine;
    if (isMockMode) {
      baseEngine = new MockAIEngine();
    } else {
      baseEngine = new GeminiAIEngine();
    }
    return new CachingAIEngineDecorator(baseEngine);
  }
}
```

2. Structural: Decorator Pattern

Description: The Decorator pattern attaches new behaviors to objects dynamically by wrapping them inside a structural class that maintains the same interface. We wrapped our base AIEngine inside a CachingAIEngineDecorator. The decorator intercepts call executions to check the database cache (ai_summaries table) first. If cached, it immediately serves the response, saving Gemini API network transactions.

Decorator Pipeline Diagram:



Implementation File Path:

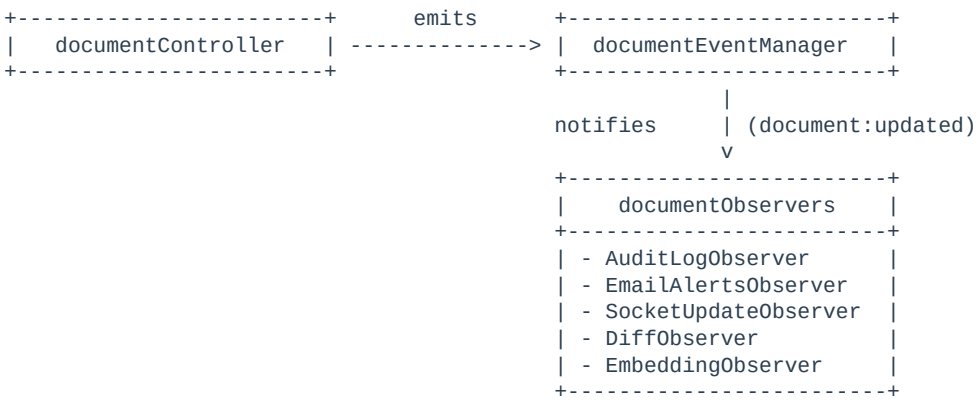
server/src/services/ai/CachingAIEngineDecorator.js

```
class CachingAIEngineDecorator extends AIEngine {
  constructor(baseEngine) {
    super();
    this.baseEngine = baseEngine;
  }

  async generateChangeSummary(versionDiffId, changes, documentTitle) {
    const cached = await pool.query('SELECT summary_text FROM ai_summaries WHERE...');
    if (cached.rows.length > 0) return cached.rows[0].summary_text;

    const result = await this.baseEngine.generateChangeSummary(versionDiffId, changes, documentTitle);
    await pool.query('INSERT INTO ai_summaries...');
    return result.text;
  }
}
```

3. Behavioral: Observer Pattern

Description: The Observer pattern defines a subscription model where a subject notifies observers of state alterations. In the legacy code, documentController.js was heavily coupled with 5 post-upload side effects (Auditing, Email Notifications, Socket Broadcasting, Visual Diffing, and Vector Embeddings). We decoupled these into isolated listeners registered to a central subject documentEventManager.

Event-Driven Observer Diagram:



Implementation Code: Event Emission (Publisher)

server/src/controllers/documentController.js

```
// Inside uploadNewVersion controller
await dbClient.query('COMMIT');

documentEventManager.emit('document:updated', {
  req,
  userId: req.user.id,
  docId,
  versionId,
  s3Key,
  nextVersion,
  fileBuffer: req.file.buffer
});
```

Implementation Code: Event Listeners (Observers)

server/src/services/observers/documentObservers.js

```
// Registering Email Alerts Observer
documentEventManager.on('document:updated', async (data) => {
  const { docId, nextVersion, versionId, userId } = data;
  if (nextVersion <= 1) return;
  await emailQueue.add('sendEmailAlerts', { documentId: docId, ... });
});

// Registering S3 Vector Embedding Observer
documentEventManager.on('document:updated', async (data) => {
  const { docId, versionId, fileBuffer } = data;
  const parser = new PDFParse({ data: fileBuffer });
  const result = await parser.getText();
  await storeEmbeddings(docId, versionId, result.pages...);
});
```

Conclusion & Reference Verification

All refactored patterns were compiled and syntax-verified locally using node --check. The structural refactoring maintains 100% backward-compatibility for LivePDF. API endpoints behave exactly as before, but the code architecture is clean, decoupled, and optimized for resource handling.